

Equations for Étendue

E Spillar

May 25, 2026

Contents

1	Symbols	1
2	Derivation of Search Rate	2
3	Conventional Units: Magnitudes	3
4	Should a Volume Search Sensor be Near or Far to Targets?	3
5	Track Before Detect	3
5.1	Definitions	3
5.2	Quantities	4
5.3	FAXT	4

1 Symbols

The units are a bit of a mess here; to make the equations a bit more transparent we work in photons, seconds, square meters, and steradians, so conversions from magnitudes will be necessary in places given common practice.

- α is the brightness of the target being sought in units commensurate with photons/second/meter².
- β is the brightness of the sky (photons/second/meter²/steradian).
- γ will be the SNR ratio.
- Ω is the area of the FOV of the sensor/optical system (steradians²).
- ω is the area of the FOV of a pixel (steradians²) or the seeing disk or the diffraction disk, whichever is largest. Basically this is the area over which the light is spread and needs to compete with the background.
- A is the effective collecting area (meter²).
- T is the integration time (seconds.) This may be the result of processing n shorter integrations t .
- g is used in some places as the limiting flux in photons/second/meter².
- η is the net quantum efficiency of the whole system
- f is the fraction of light from an object captured in the photometry aperture
- C is the calibration which gives the number of photons per second per square meter for a 0 magnitude object for the band being used.

2 Derivation of Search Rate

The signal from an object is $S \propto \alpha A T \eta f$. Assuming we are background limited, the noise is $N \propto \sqrt{\beta A T \eta \omega}$. Putting things together

$$SNR^2 = \gamma^2 = \frac{\alpha^2 A^2 T^2 \eta^2 f^2}{\beta A T \eta \omega} \quad (1)$$

$$\gamma^2 = \frac{\alpha^2 A T \eta f^2}{\beta \omega} \quad (2)$$

so

$$\boxed{T = \frac{\gamma^2 \beta \omega}{\alpha^2 A \eta f^2}} \quad (3)$$

This the “dwell time” required to reach the given SNR for the given object (assuming BLIP).

To get the search rate, multiply by the total area surveyed:

$$\boxed{SR = \frac{\Omega}{T} = \alpha \left(\frac{\alpha}{\gamma} \right)^2 \frac{A \Omega}{\omega} \frac{\eta f^2}{\beta}} \quad (4)$$

which has units commensurate with steradians/second.

3 Conventional Units: Magnitudes

Typically, we will be numbers in magnitudes in a selected band; this will be magnitude for a source or magnitude per square arc second for a background number. For each band there is a calibration C_b which gives the number of photons per second per square meter for the filter band b in question so the flux f is given in terms of the magnitude m

$$m = -2.5 \log_{10}(f/C_b) \quad (5)$$

and

$$f = C_b 10^{-0.4m} \quad (6)$$

4 Should a Volume Search Sensor be Near or Far to Targets?

Thinking naively, if you are further away from targets, you will be able to cover more area or volume much faster, so perhaps that's advantageous? There are some subtleties here, but let's assume that the limiting factor is reaching the faintest outer shell of the volume to be surveilled. Fixing the scanning of this shell as a requirement, and fixing the FOV of the telescope, the area of the FOV projected onto the shell $A_{FOV} \propto d^2$; so assuming a fixed dwell time the search rate also goes as d^2 . But the dwell time needs to increase since sources will appear fainter. The flux from the target $f \propto 1/d^2$. Assuming BLIP once again, the required integration time, or required dwell time, goes as $t \propto d^{-4}$. Hence our search rate goes as

$$\text{search rate} \propto \frac{1}{d^2} \quad (7)$$

It's best to place the sensor close to the surveilled volume.

5 Track Before Detect

5.1 Definitions

- θ is the angular size of a pixel
- n_x by n_y pixels are being explored for moving objects
- t is the time between frames frame time
- n is the maximum number of lags being calculated and we assume that we will be doing motions of up to 1 pixel per lag.
- n_{vx} and n_y are the velocities

5.2 Quantities

The maximum rate being detected is

$$\text{Angular Rate Sensitivity} = \theta/t. \quad (8)$$

The number of operations using a naive algorithm is

$$4n_x n_y n n_{vx} n_{vy}. \quad (9)$$

I am uncertain about the 4 in front, this needs to be reviewed.

If we use the FAXT algorithm, I think that the number is reduced to

$$4n_x n_y n_{vx} n_{vy} \log_2(n) \quad (10)$$

No this is too simple. I need to spend some time and read the paper in detail.
Let's go ahead and do that.

5.3 FAXT